



MRC
Biostatistics
Unit



UNIVERSITY OF
CAMBRIDGE

Adaptive Methods in Clinical Research

Some thoughts on ICH E20 Guideline for Adaptive Designs

Sofía S. Villar

MRC Biostatistics Unit, University of Cambridge

November 19th 2025

Objectives for this Recap

- Define “Adaptive Trial”: Establish a common working definition.

Identify Key Elements: Summarize the core types of adaptive elements

- The guidance highlights the need for Pre-specification, Type I error rate control, and unbiased estimation.
- Set the Stage: Ground the subsequent discussions on Response-Adaptive Randomization (RAR), Estimation after Adaptive Designs (ADs), Ethics of ADs and panel discussion.

Outline

1. Introduction and Scope
2. Advantages and Challenges of Adaptive Designs
3. Key Principles
4. Types of Adaptations
5. Special Topics and Considerations
6. Documentation

- What is the focus? **confirmatory trials** with an **adaptive design** to evaluate a treatment within the context of its overall development programme (confirm efficacy and support benefit-risk assessment).
- What is an **adaptive design (AD)** (and what is not)?

A clinical trial allowing **pre-specified**, prospectively planned modifications to one or more aspects of the trial based on interim analysis of accumulating participant data.

- **Not an AD:** unplanned modifications in response to interim results or external data.
- Focus is on interpretable results from adaptive designs (all phases), not on prescriptive methods.

Outline

1. Introduction and Scope
2. Advantages and Challenges of Adaptive Designs
3. Key Principles
4. Types of Adaptations
5. Special Topics and Considerations
6. Documentation

The Adaptive Designs Trade-Offs

- Even after careful earlier phases, there may be **residual uncertainty** before confirmatory trials.

An adaptive design is a “*safeguard against inaccurate assumptions*” as it allows to change some design elements otherwise fixed at the start of the trial.

- **Advantages:** ethical “*reduce the number of participants to an inferior control*”; efficiency “*increase power given an expected sample size* and “*improved understanding of treatment effects and decision making* (better benefit-risk profile).
- **Challenges:** Require longer development/analysis time, carry risks of operational bias (due to information leakage from interim analyses), potentially biased treatment effect estimates and reduced risk-benefit information, and need high-quality, real-time data for reliable adaptation.

“... essential to carefully justify the need to adapt the trial and potential implications of adaptations involved”

“A proposed adaptive design requires a clear and compelling justification”

- Justification should include **both** clinical and statistical considerations.
- Evaluate advantages and limitation in comparison to alternative designs (including non-adaptive ones).

To justify one must justify how the **residual uncertainty** handled with an adaptive design will be less than the **uncertainty** about the trial's ability to produce **reliable and interpretable results**. [Personal interpretation: Why adapt?]

Outline

1. Introduction and Scope
2. Advantages and Challenges of Adaptive Designs
- 3. Key Principles**
4. Types of Adaptations
5. Special Topics and Considerations
6. Documentation

- **Principles:** = those characteristics critical to ensuring reliability and interpretability of results across all types of adaptations.
 1. Adequacy within the Development Program
 2. Adequacy of Trial Planning
 3. Limiting the Chances of Erroneous Conclusions
 4. Reliability of Estimation
 5. Maintenance of Trial Integrity

Adequacy within the Development Program

- *Confirmatory trials are informed by results from a stepwise exploratory program.*
- Such earlier phase studies seek to reduce uncertainty on multiple aspects such as:
 - ▶ Dose-response relationship (favorable and unfavorable effects)
 - ▶ Identify the appropriate patient population
 - ▶ Inform endpoint choice (clinically meaningful)
- *“The number and complexity of adaptations at the confirmatory stage should generally be limited.” [Circle back to this in RAR]*
- Adaptive elements in a confirmatory stage must **selectively target residual uncertainty after exploratory stages**, limiting the number of adaptive elements to maintain overall design interpretability. [Personal interpretation: What to adapt?]

Adequacy of Trial Planning

- For an AD there are specific additional elements to be prespecified (compared to a non-adaptive trial). These include:
 - ▶ Number and timing of interim analysis (**how often and when?**)
 - ▶ Type of adaptation(s) and anticipated rule for the adaptation (binding or not) and impact on inference (**what and how?**)
 - ▶ Statistical methods for interim and final primary analysis (aligned to estimand)
 - ▶ Approaches to maintain integrity
 - ▶ Discussions with the Independent Data Monitoring Committee (IDMC) to ensure they are prepared to review interim results and make adaptation recommendations while overseeing safety

Limiting Chances of Erroneous Conclusions

- *An essential element of regulatory decision-making is controlling chances of false efficacy conclusions.*
- However, it is also important to limit erroneous conclusions about safety and benefit-risk profile of a new treatment as well.
- ICHE9: Common approach to limit type I error is based on *frequentist* methods. Other approaches (*Bayesian*) may be appropriate “*when reasons for use are clear and conclusions sufficiently robust*” [Circle back in RAR]
- Adaptations can affect the chance of a false negative efficacy conclusion so a design-specific hypothesis test is needed. (Also, adequate power should be established).

- *“reliable estimation of treatment effects for the primary efficacy endpoint and other key outcomes is important to facilitate benefit-risk assessment and regulatory decision-making”*
- Estimation after an AD should recognize the effect of the AD on the resulting (point and confidence) estimates. This includes bias, variability, coverage of confidence intervals. [**see the lectures on estimation for ADs**]
- Also important to ensure reliable adaptation by careful consideration of **interim analysis estimates** and their expected bias/variability.

Maintenance of Trial Integrity

- Trial integrity requires both **statistical validity** and **appropriate operational execution** to ensure objectives are achieved reliably, ethically, and promptly. Assessing how adaptations can impact trial conduct is essential.
- “...many types of ADs need some level of access to unblinded interim results”
- Knowledge by the sponsor, investigators or trial participants about: individual assignments, accumulating data (summaries) and/or trial adaptations triggered can introduce subtle changes in trial conduct that will be impossible to adjust for.
- **Recommendations:** Implement changes under strict blinding (where possible); an expert IDMC, use a wide range of estimates to trigger a specific adaptation; and ensure sufficiently large stage size to assess heterogeneity.

Outline

1. Introduction and Scope
2. Advantages and Challenges of Adaptive Designs
3. Key Principles
- 4. Types of Adaptations**
5. Special Topics and Considerations
6. Documentation

4.1 Early Trial Stopping

- “...*accruing (efficacy) data can provide information that makes it no longer appropriate to continue the trial*”
- Group Sequential Design (GSD): prospectively planned sequential analysis of the accumulating unblinded data with appropriate stopping boundaries (i.e. that control for type I error). Efficacy and/or futility stopping. Nonbinding futility rules.

Adaptation targets the total sample size, no longer fixed.

- Pre-specify and justify interim analysis timing based on statistical factors (persuasiveness, stopping probability, risk assessment); allow minor, justified operational deviations (e.g., IDMC schedules) where the actual sample size may differ.
- After an early stop, the time lag needs reporting two treatment effect estimates (interim and final) to monitor for divergence and ensure a consistent basis for benefit-risk assessment.

4.2 Sample Size Adaptation

- “... a considerable degree of uncertainty might exist in the parameter assumptions that affect the (initial) sample size”
- Sample Size Adaptation (SSA) addresses justified residual uncertainty (in either nuisance parameters or treatment effect size) to ensure an adequately powered trial.

Adaptation targets the total sample size, no longer fixed.

- Pre-specification and justification of the **minimum** (sufficient information for benefit-risk assessment) and **maximum** (practical considerations like recruitment capacity) trial sizes, the **adaptation rule**, and the statistical **final analysis** method.
- Blinded estimation for nuisance parameter (to protect integrity) and a testing approach that strictly controls the Type I Error rate. Unblinded estimation for effects (through IDMC), clinical relevant effect sizes, SSA methods that do not strictly enforce the anticipated rule (e.g., combining information across stages).

4.3 Population Selection

- “...remaining uncertainty about the patient population who should be treated with anew treatment”

Uncertainty in population definition implies: either (i) studying the overall population risks insufficient power if the benefit is only in the subgroup or (ii) restricting the indication to a subgroup when the overall population also benefits.

- **Adaptive Enrichment:** Start with enrollment across the overall population and use an interim analysis to decide whether to continue in the overall group or focus only on a pre-defined subgroup. Alternative to a multiple testing approach.

Adaptation targets the target population, no longer fixed.

- Type I error control when combining stage data.
Pre-specification: candidate population(s) for selection, the final inference population, and the adaptation rule. For non-binary features, pre-specify and justify all subpopulation thresholds.

4.4 Treatment Selection

- “...*uncertainty remaining about the benefit-risk profile of two likely efficacious doses of a drug*”
- A two-stage decision, where first stage doses are compared against a control (to confirm efficacy) and then one is selected to go to a second stage (to confirm efficacy and benefit-risk of the selected dose).

Adaptation here targets the specific (number of) treatments included in the trial, no longer fixed.

- Pre-specification of treatments, decision rules, and resource implications (randomization/sample size). Specific methodology to control Type I error; recommended flexibility in the adaptation rule and final estimates need adjusting for bias.

4.5 Adaptation to Participant Allocation

- “...typically allocated to treatment arms according to fixed randomization probabilities”
- Response-adaptive randomization: makes treatment assignment depend of accumulating outcome data.

Adaptation here targets the allocation ratio per arm, which is no longer fixed.

- Motivation: “favor best performing treatments” (What is the residual uncertainty here: effect size?)
- Challenges: bias, type I error control, effect of time trends, less information on benefit-risk profile. Integrity requires no sponsor involvement. Timely and high quality interim data.
- Recommendations: stratified analysis, randomization based analysis. Avoid deterministic procedures.

Outline

1. Introduction and Scope
2. Advantages and Challenges of Adaptive Designs
3. Key Principles
4. Types of Adaptations
5. Special Topics and Considerations
6. Documentation

5.1 Further considerations on Data Monitoring

- “...*further considerations for ADs with unblinded interim analysis of accumulating data*”
- The expert IDMC (including a specialized statistician and all expertise needed to make adaptation recommendations) must operate under a detailed charter and ensure upfront alignment with the sponsor on all adaptive rules, in addition to their core safety role.
- An external, independent group with the necessary expertise must be established to conduct unblinded analyses, holding sole access to interim data for the IDMC. Adaptations must be implementable by the sponsor without unblinded data access.
- Any proposed access by sponsor to unblinded data requires compelling rationale, rigorous integrity measures (firewalls), and detailed documentation for regulatory review.

5.2 Planning, Conducting and Reporting Simulation Studies

- “*..simulations play an important role planning an AD and for understanding their operating characteristics*”

Simulations are repeated executions of hypothetical trials under various scenarios.

- Important to clearly define objectives directly related to decisions that will be made from the simulation study. Decisions include: number and timing of interim analyses, adaptation types, analysis and estimation methods.
- Include a **benchmark** design to quantify trade-offs and inform the final choice. Pre-define and assess a broad set of operating characteristics (Type I error, power, bias, expected sample size, and confidence interval coverage) aligned with study's objectives. Beyond averages, evaluate the variability and distribution of results (e.g., sample size distribution).

5.2 Planning, Conducting and Reporting Simulation Studies (Cont'd)

- Simulation scenarios must cover the **plausible range of assumptions** (treatment effects, nuisance parameters, and operational factors like recruitment rates and dropout rates) to ensure a robust assessment of the AD performance.
- Adequacy of all assumptions must be clinically and statistically justified using supporting evidence (e.g., previous trial data, publications). If using a grid of assumptions, justify that this covers all plausible scenarios and that it is sufficiently fine to provide a reliable estimate of the operating characteristics.
- Clear specification of data-generating process (particularly when complex model fit involved).
- Use sufficient repetitions (e.g., 100,000+ for Type I error precision); Document all algorithms and ensure random numbers are reproducible (via a documented seed).

5.2 Planning, Conducting and Reporting Simulation Studies (Cont'd)

Simulation Documentation (Regulatory Submission)

- A comprehensive report of the simulation (design, results and conclusions) establishing explicit links between clinical assumptions and statistical results.
- The report must cover: the key questions addressed, the full range of design and analysis options and operating characteristics assessed (including Type I error and bias), and all supporting knowledge (references).
- Reproducibility: data-generating process, number of repetitions, and include the simulation software/code (with comments and execution instructions).
- Include a summary that discusses overall results, interpretations, and conclusions (including limitations and clinical discussion) of the proposed AD's performance.

5.3 Adaptive Designs Using Bayesian Methods¹

- ¹ *“This section is not fully harmonized”*
- Bayesian methods are considered when their use is clearly justified and the resulting conclusions are sufficiently robust (as per ICH E9).
- *“Bayesian methods refer to a wide range of statistical approaches that combine prior probability distribution with current trial data to obtain a posterior distribution for a quantity of interest.”* Such posterior can inform adaptations.
- For such designs, the expectations for operating characteristics (power, expected sample size) remain the same as for non-Bayesian adaptive designs. Set all elements (prior distribution, decision criteria, and adaptation rules) to achieve other targets while strictly controlling the Type I error probability.

5.3 Adaptive Designs Using Bayesian Methods¹

- ¹ *“This section is not fully harmonized”*
- Special case: ADs with borrowing of external information via an informative prior and a primary analysis based on the posterior of the same estimand of interest.
- This justification must address the feasibility of running a fully powered trial without borrowing and support the relevance and quality of the external data used.
- Misspecification of the informative prior introduces significant uncertainty and can lead to a lack of control of the false positive probability.

5.4 Adaptive Designs in Time-to-Event (TTE) Settings

- In TTE trials, power depends on the number of events, not participants. Interim analyses and adaptations are therefore event-count driven.
- In SSA the target event count is increased via longer follow-up, increased enrollment, or both. This adaptation requires ensuring sufficient long-term data for benefit-risk assessment.
- TTE trials violate the ideal of independent stages because participants enrolled early may have their event in a later stage. Using intermediate data (e.g., secondary endpoints or event-free status) from these participants to inform adaptations can inflate the Type I error probability.
- Type I Error control requires specialized TTE methodology: use a non-deviating rule (primary endpoint only) or define stages with fixed follow-up/event targets.

5.5 Adaptive Designs in Exploratory Trials

- *“If a trial may be intended to confirm efficacy and support benefit-risk assessment, it is critical that the principles in ICHE20 are followed.”*
- Exploratory trials often have additional considerations because their findings will be confirmed later.
- For instance, they may only need to describe general principles for adaptations rather than specific rules.
- Due to the multidimensional decisions involved (like selecting a dose), sponsors may require greater internal input, which demands careful balancing between the need for sponsor involvement and the strict necessity to maintain trial integrity by limiting access to unblinded results.

5.6 Operational Considerations

- Trial Integrity: Operational planning to minimize information leakage from interim analyses.
- Consent forms must cover all potential adaptive changes and assure participants of continued safety.
- ADs require **specialized infrastructure**, typically involving integrated Interactive Response Technology (IRT) systems. These systems must have pre-specified algorithms to handle foreseeable adaptations (like changes to treatment arms or randomization ratios) with minimal sponsor intervention.
- ADs often strain the drug supply chain due to lead times for manufacturing. Sponsors must use simulations to proactively plan the drug supply and distribution, especially for multi-regional trials, to prevent treatment delays.
- Quality interim data via timely validation and formal database lock.

Outline

1. Introduction and Scope
2. Advantages and Challenges of Adaptive Designs
3. Key Principles
4. Types of Adaptations
5. Special Topics and Considerations
6. Documentation

6.1 Documentation Prior to Conducting a Confirmatory Trial with an Adaptive Design

1. Rationale: Justify the AD clinically and statistically, discussing advantages and limitations compared to non-adaptive alternatives to address any added uncertainty.
2. Detail aspects to be modified, interim timing, and the adaptation rules (with min/max sizes and candidate estimand definitions).
3. Statistical Analysis: Detail methods for interim results, the approach for primary/secondary analysis (testing, estimation, uncertainty measures), and sensitivity analyses.
4. Define committee roles and responsibilities for analysis and decision-making.
5. Describe access control for unblinded data and record-keeping.
6. Document key Operating Characteristics; include the full simulation report if studies were critical.

6.2 Documentation to Include in a Marketing Application After a Completed Confirmatory Trial with an Adaptive Design

- The application must include all pre-specified prospective plans (Section 6.1) alongside details on how the adaptation was implemented—including the actual timing and results of interim analyses, any notable heterogeneity between trial stages, and whether the anticipated adaptation rules were followed.
- Deviations must be rationalized and their impact assessed. Provide compliance records (including audit results) confirming data access and integrity processes.
- The submission must include records of IDMC deliberations (e.g., meeting minutes) related to adaptation decisions. Finally, the analysis and reporting of final results must appropriately account for the adaptive design.